



ADI *Power*

Power Management Fundamentals II Session 3: Demystify SEPIC, Ćuk and Inverter Topologies

Piyu Dhaker

Powering Up – Module Overview

Session 1

LDOs & Bucks

LDO
Regulators

Buck
Regulators

Takeaway

A Deeper Look
into the workings
of LDOs & Bucks

Session 2

Boost &
Buck-Boost

Boost
Regulators

Buck-Boost
Regulators

Takeaway

A Deeper Look into
the workings of
Boost & Buck-Boost

Session 3

Sepic, Ćuk &
Inverters

Sepic/Ćuk

Inverting
Regulators

Takeaway

A Deeper Look into
the workings of
Sepic/Ćuk &
Inverters

Session 4

Isolated
Converters

Flyback
Converters

Forward
Converters

Takeaway

A Deeper Look at
Isolated Converters

Session 5

Power Path, Hot-swap
& Surge-Stopper

Power Path

Hot-Swap

Surge-
Stopper

PoE

Takeaway

A Deeper Look into
Power Input
Circuits

Session 6

Layout
Considerations

Layout

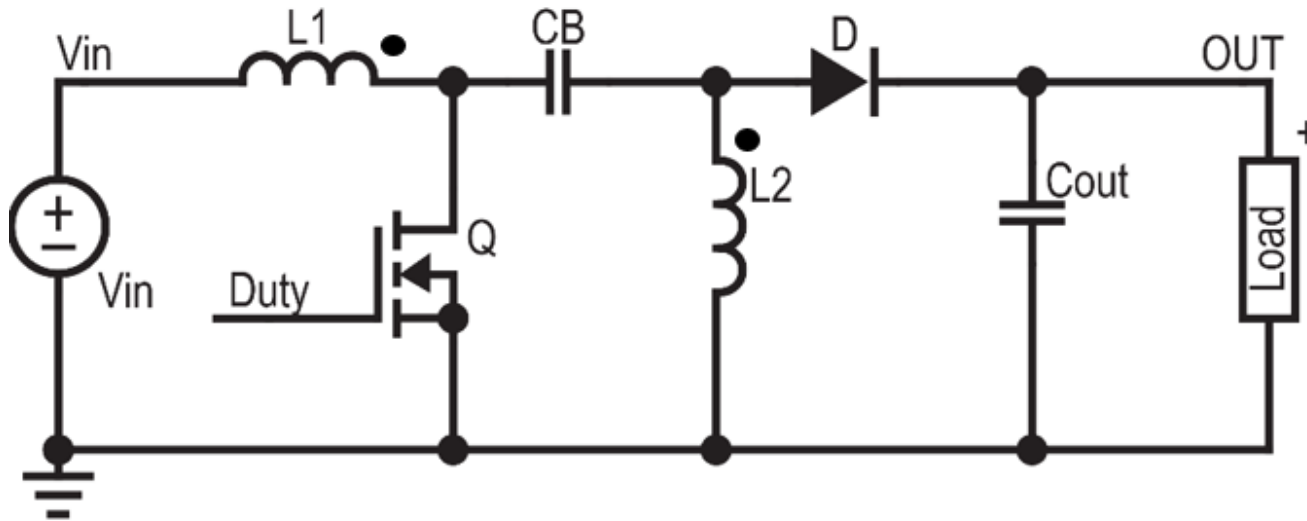
Takeaway

A Deeper Look into
Layout
Considerations

Agenda

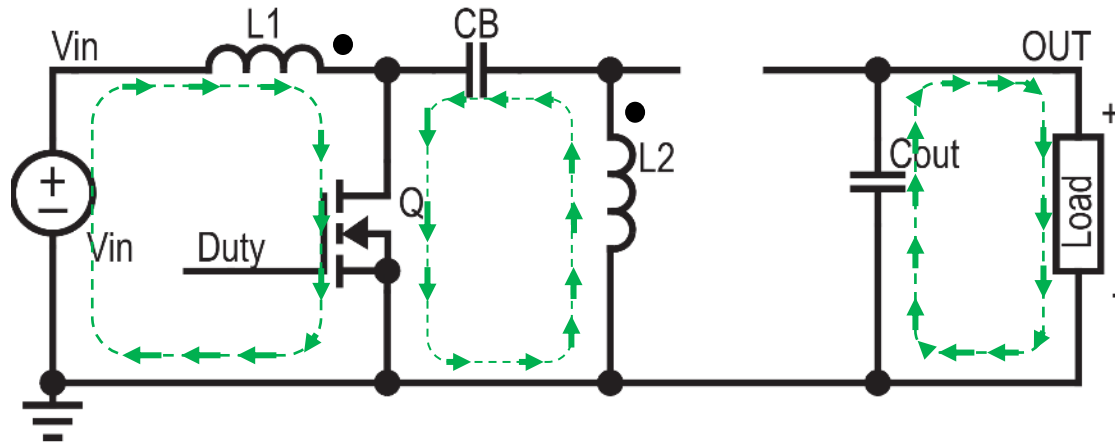
- 1 Sepic Regulators
- 2 Ćuk Regulators
- 3 Inverting Regulators

SEPIC Converters



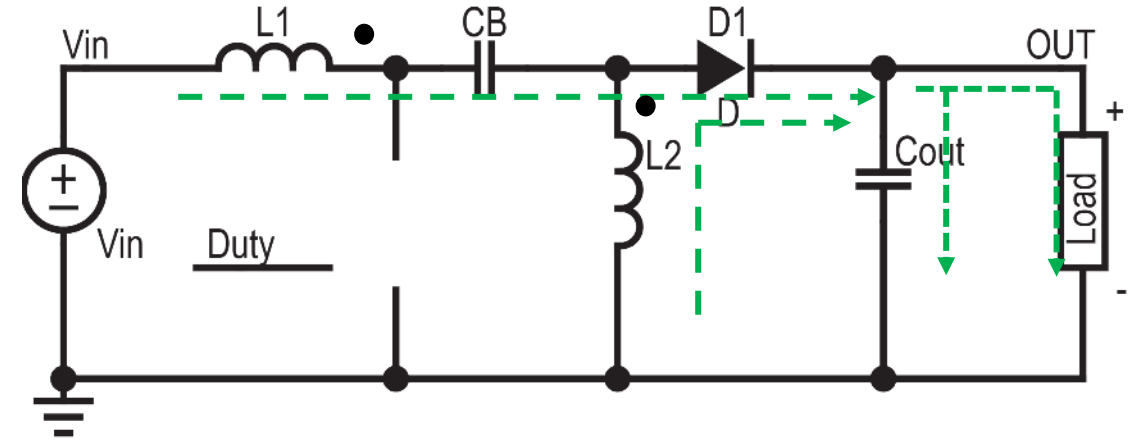
- **S**ingle-**E**nded **P**rimary **I**nductance Converter
- $V_{OUT} < V_{IN}$, $V_{OUT} = V_{IN}$, $V_{OUT} > V_{IN}$

SEPIC Converters



ON-Phase:

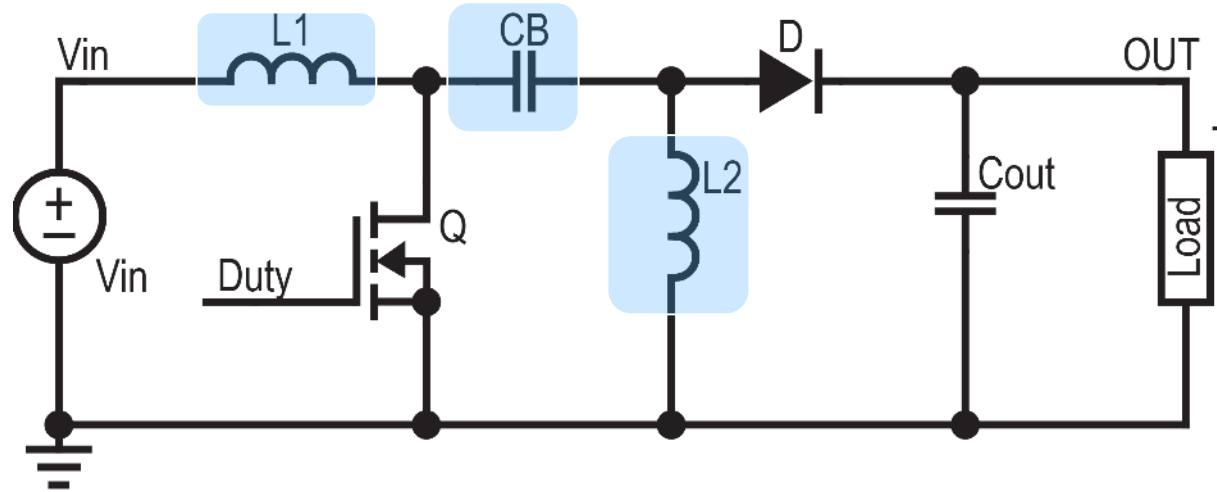
- Primary switch ON and conducts current to GND.
- Current builds up in L_1 and L_2 .
- During this time, output capacitor, C_o provides all load current



OFF-Phase:

- Primary switch is OFF.
- Current in inductors L_1 and L_2 provide output current and re-charge output capacitor, C_{out}

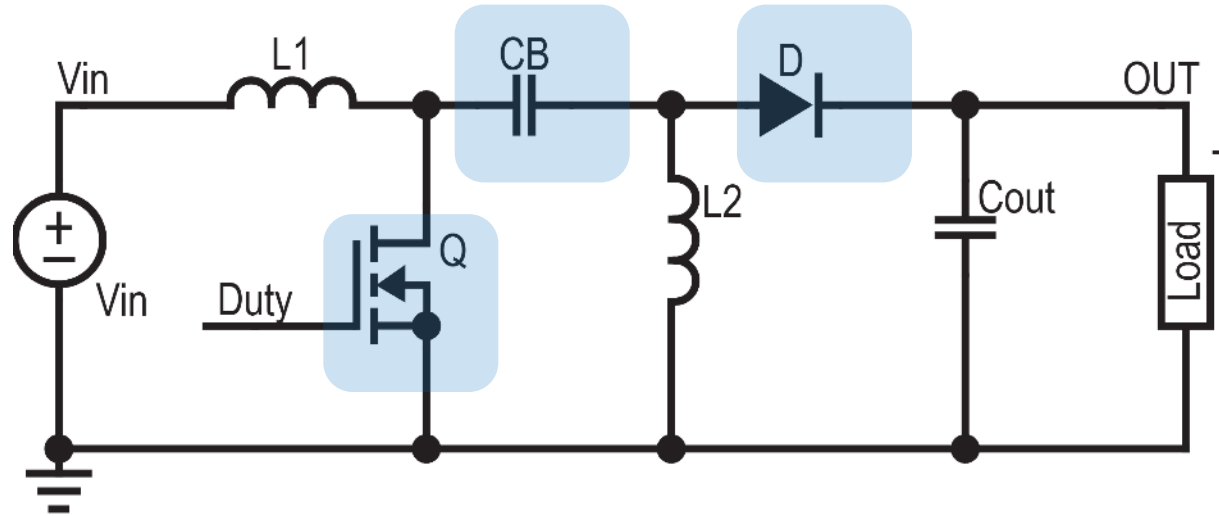
SEPIC Converters



SEPIC converters

- Inductors can be Coupled or Uncoupled
- Cap C_B provides DC isolation when the converter is OFF.
 - This is useful if you want a Boost with output disconnect

SEPIC Converters



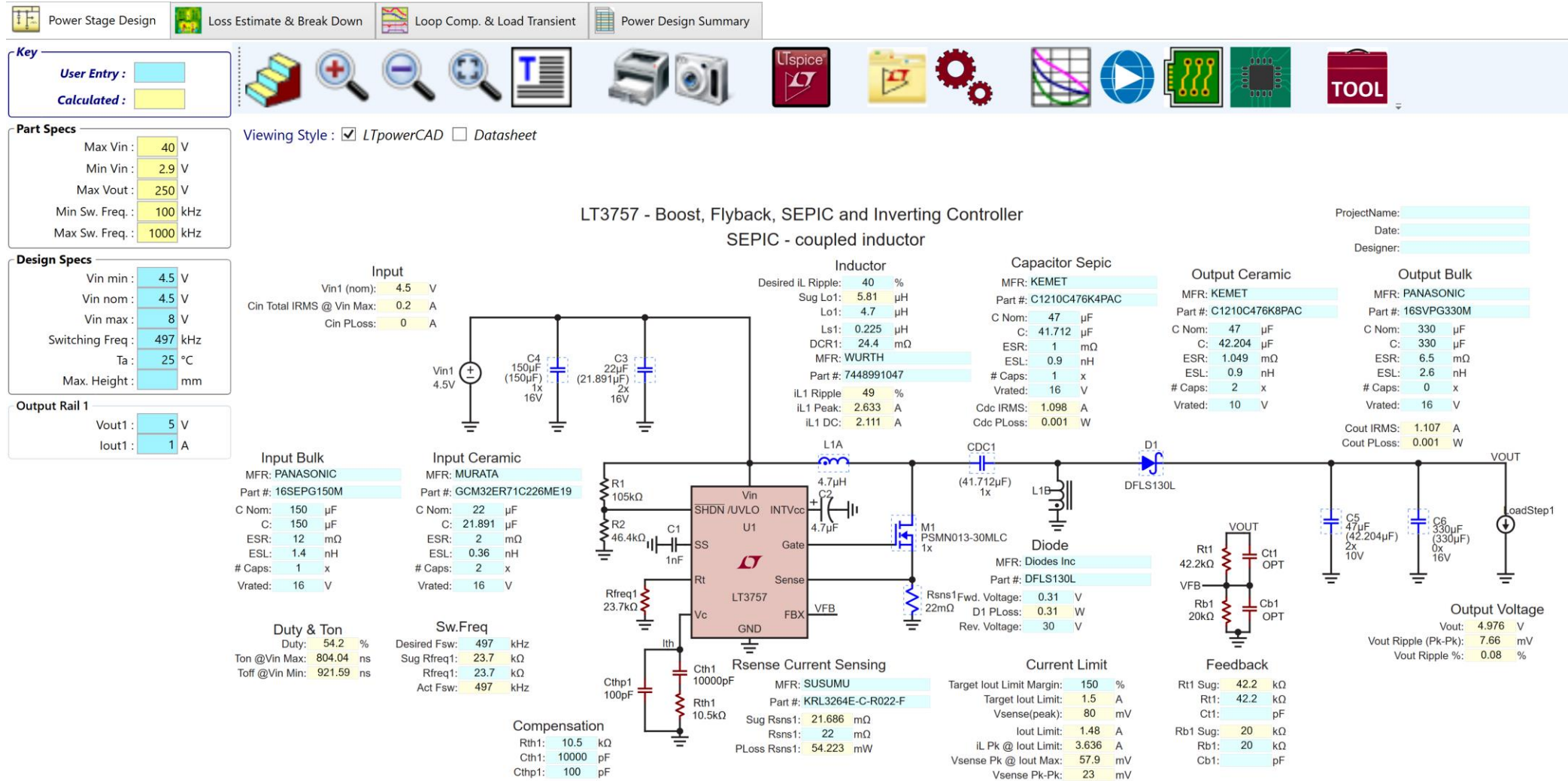
SEPIC converters

- Voltage ratings:
 - Primary Switch will see $(V_{IN} + V_{OUT})$
 - Coupling cap will see V_{IN}
 - Diode will see $(V_{IN} + V_{OUT})$
- Current Ratings (average):
 - Primary Switch will see $(I_{IN} + I_{OUT})$
 - Diode will see I_{OUT}

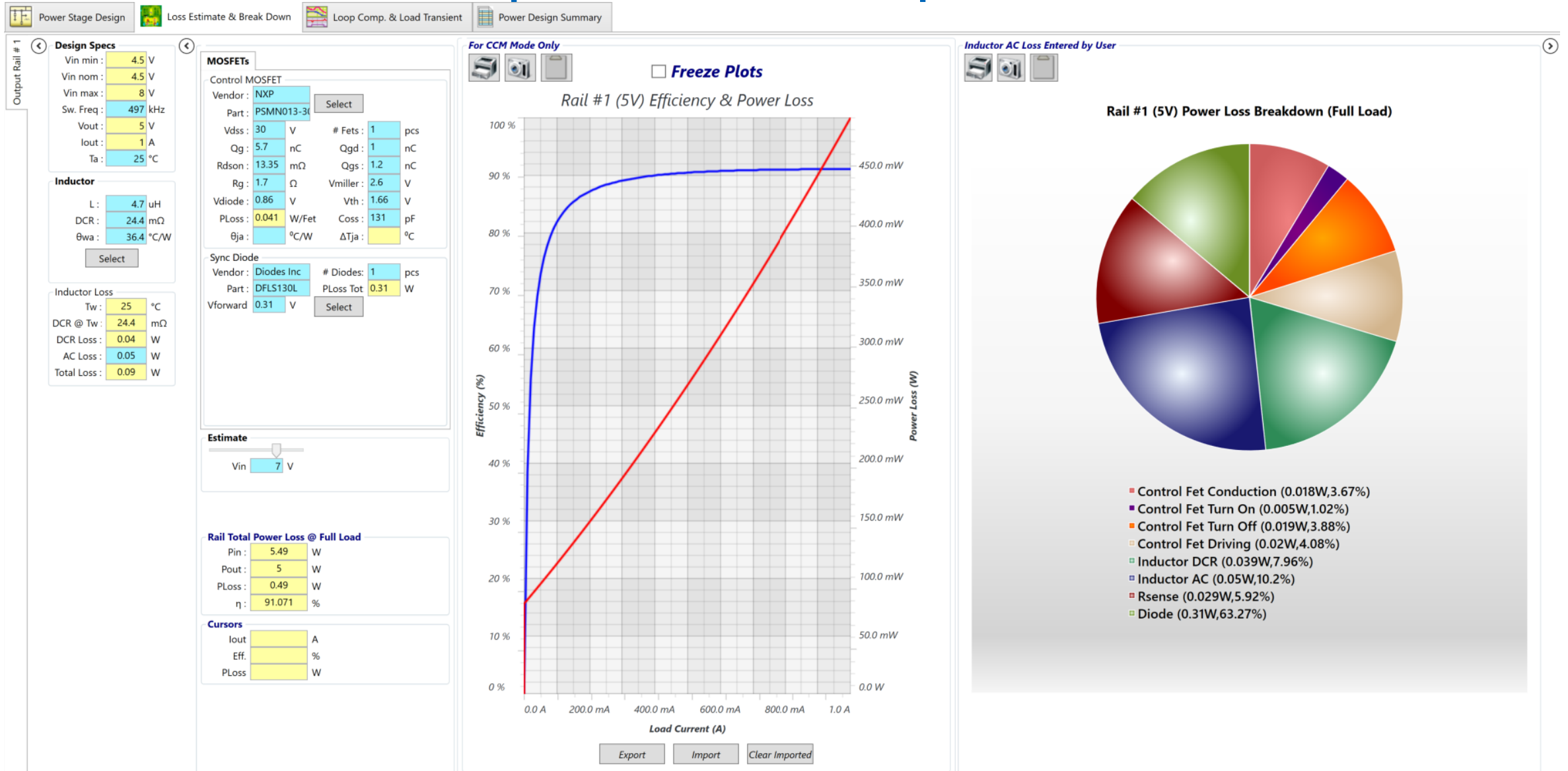
SEPIC Converters

- LT3757 chosen as example
 - V_{in} : 4.5V to 8V
 - V_{out} : 5V, 1A at 500KHz
 - LTPowerCAD example
 - LTSPICE example

SEPIC Converter: LTpowerCAD Example

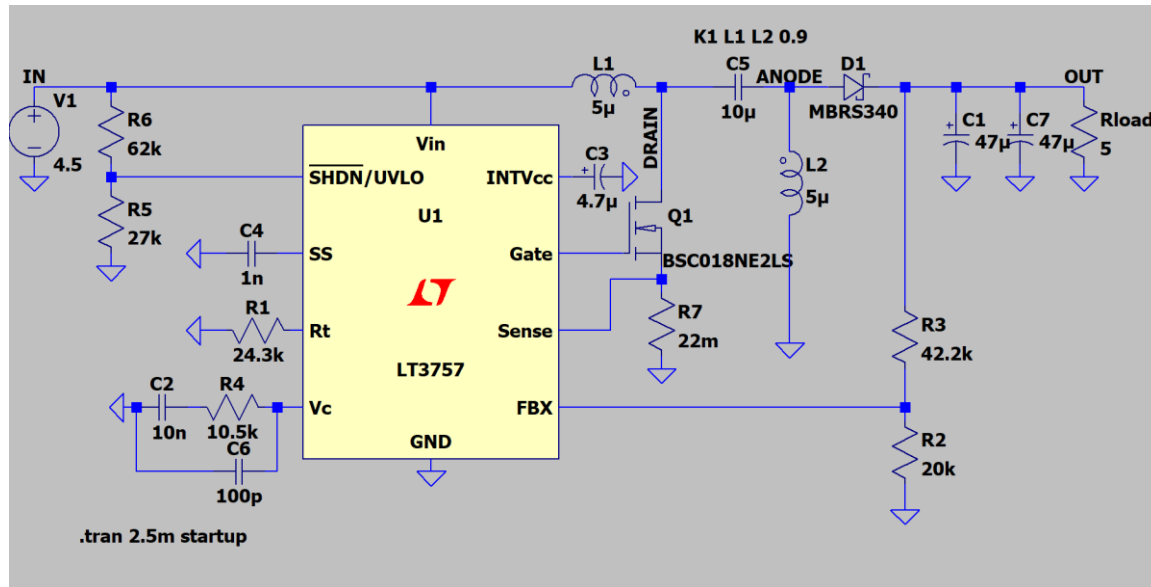


SEPIC Converter: LTpowerCAD Example

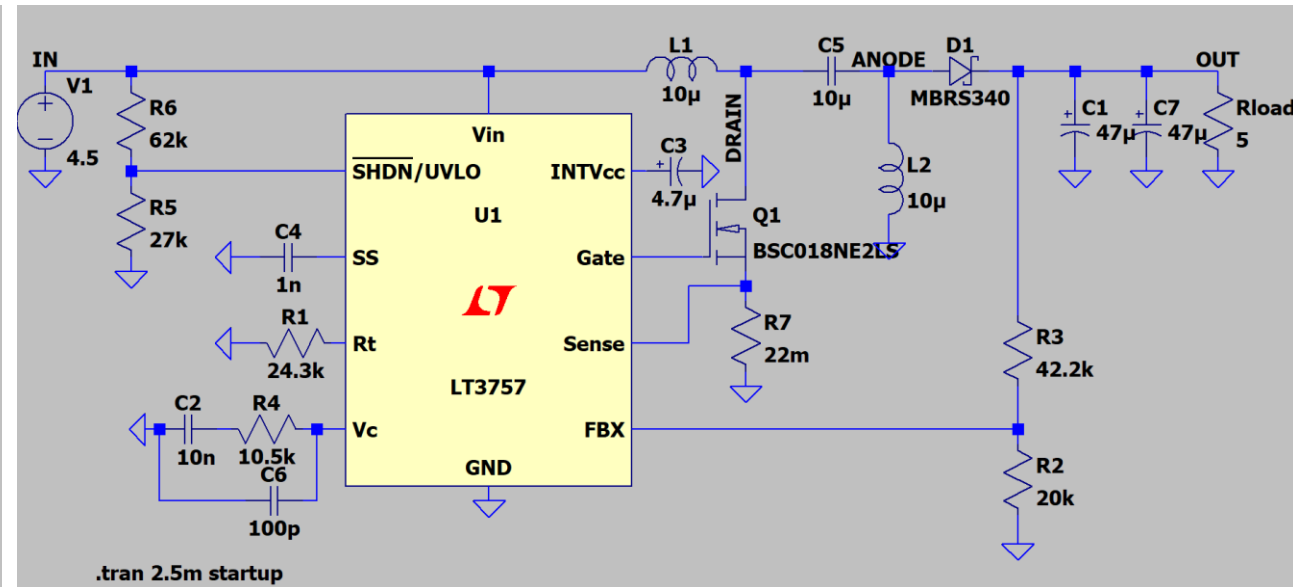


SEPIC Converter LTSPICE example

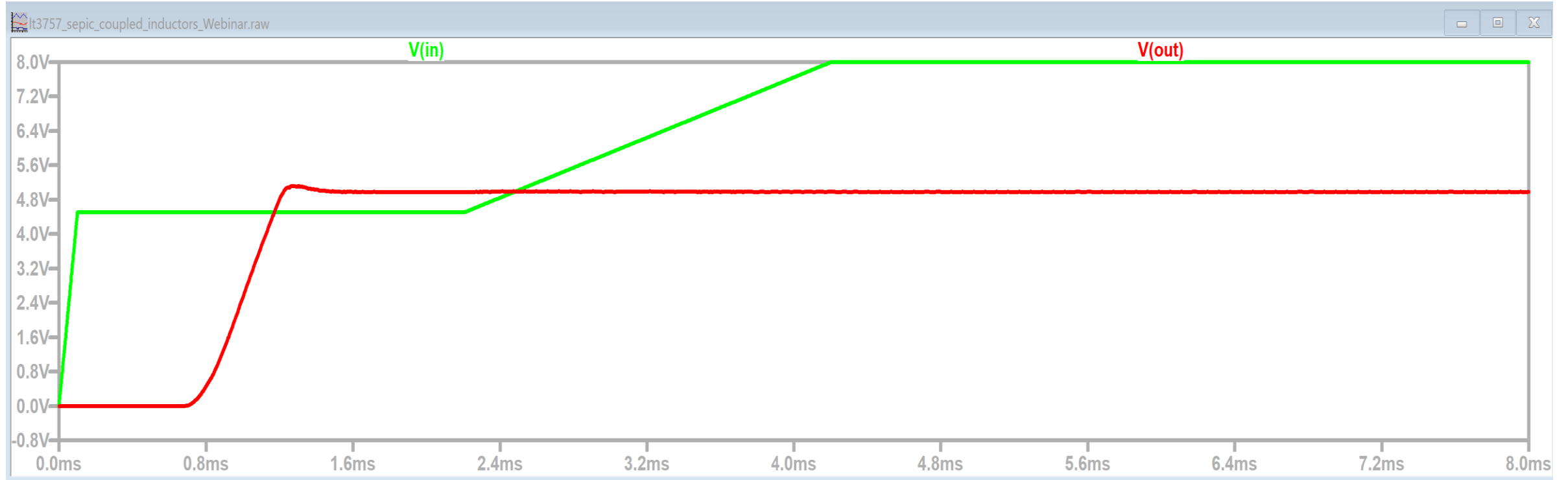
Coupled Inductor



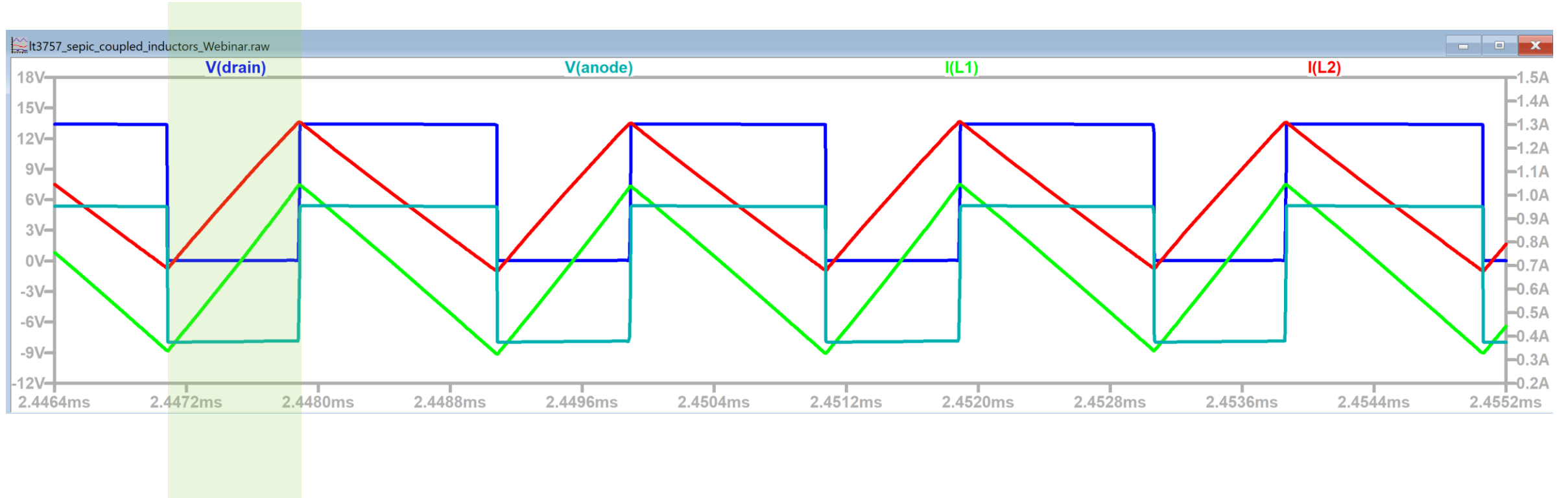
Uncoupled Inductor



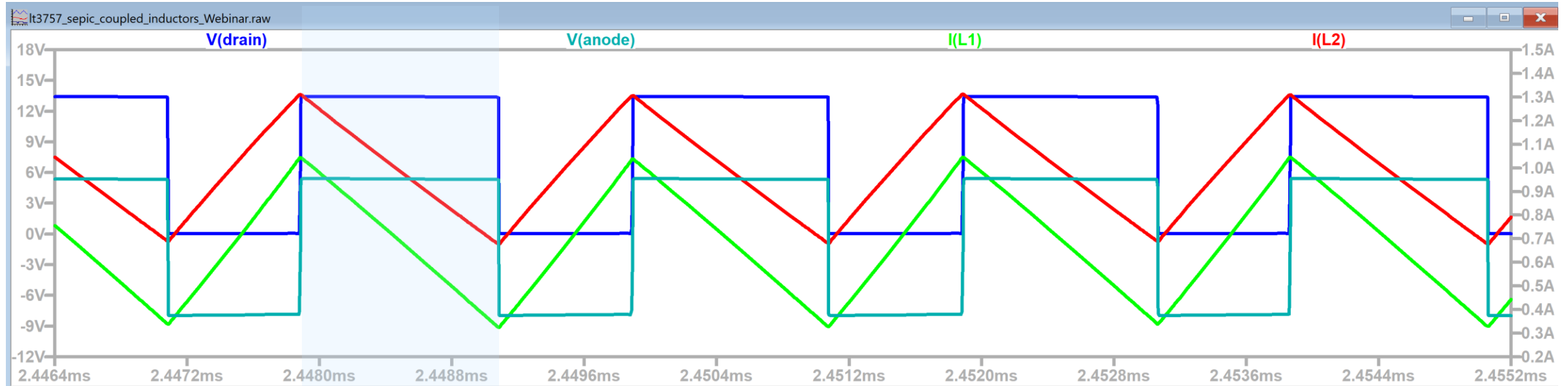
SEPIC Converter LTSPICE example



SEPIC Converter: LTSPICE Example



SEPIC Converter: LTSPICE Example



SEPIC Converters

Advantages

- Can regulate with V_{IN} above or below V_{OUT}
- Simpler circuit than a 4-switch Buck-Boost
- Provides output disconnect in all modes

Disadvantages

- Switch ratings must be greater than for 4-switch Buck-Boost
- Lower efficiency
- Requires 2 inductors

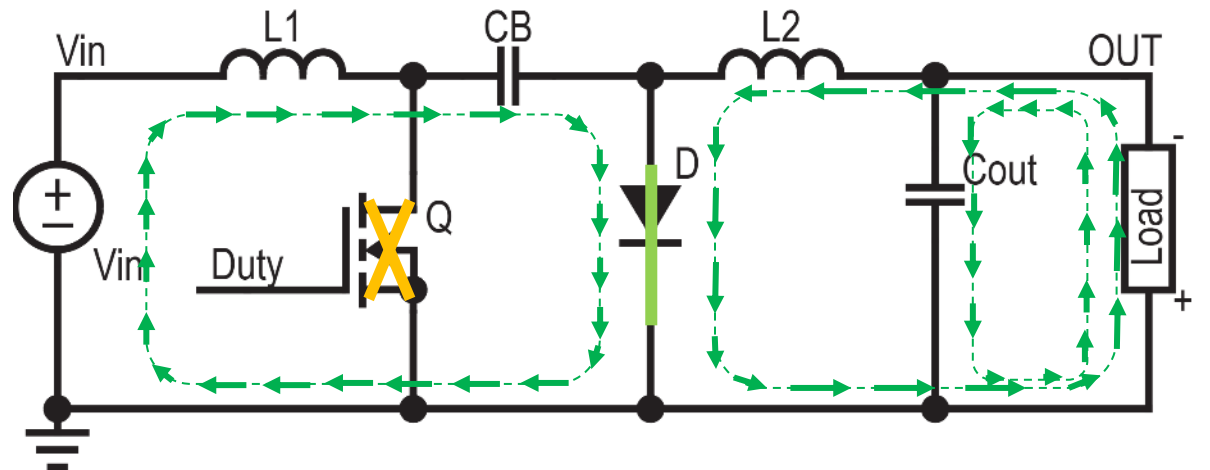
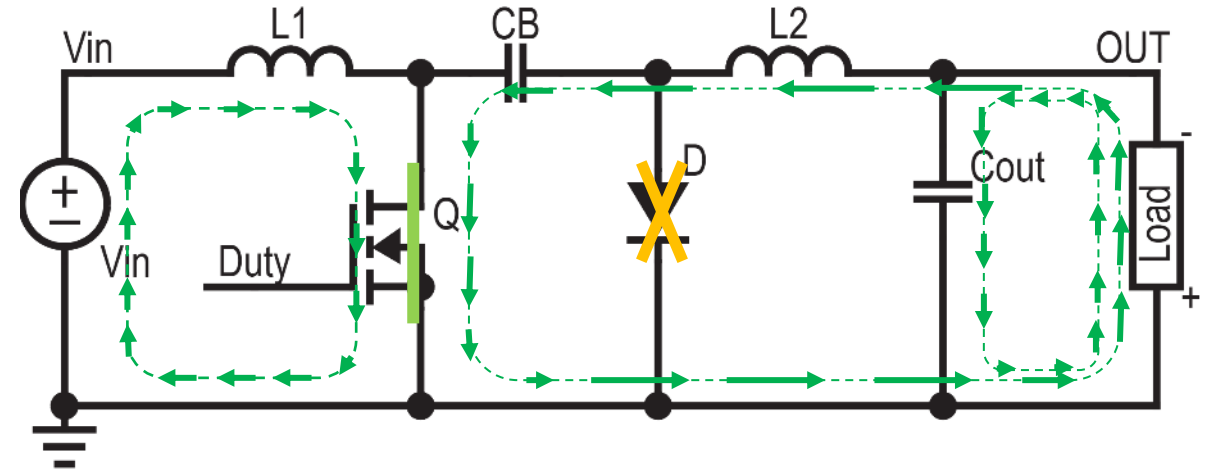
Agenda

- 1 Sepic Regulators
- 2 Ćuk Regulators
- 3 Inverting Regulators

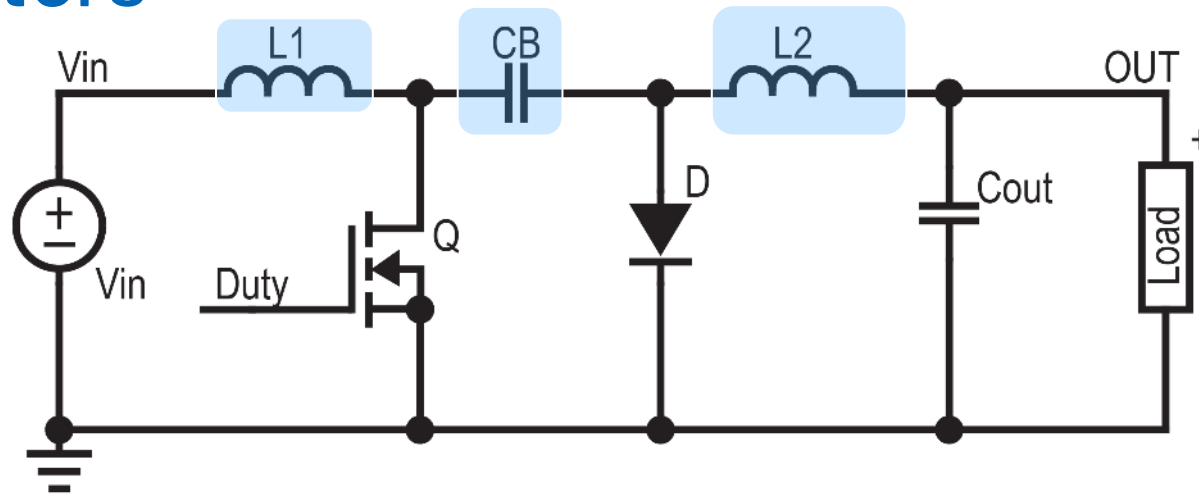
Ćuk Converters

Ćuk converters

- Named after Slobodan Ćuk, the CalTech Professor who created the topology
- V_{IN} is positive, V_{OUT} is negative
- ON-Phase: Primary switch conducts current to GND. Current builds up in L_1 and L_2 .
- OFF-Phase: Primary switch is OFF. Current in inductors L_1 and L_2 provide output current and re-charge C_{OUT}



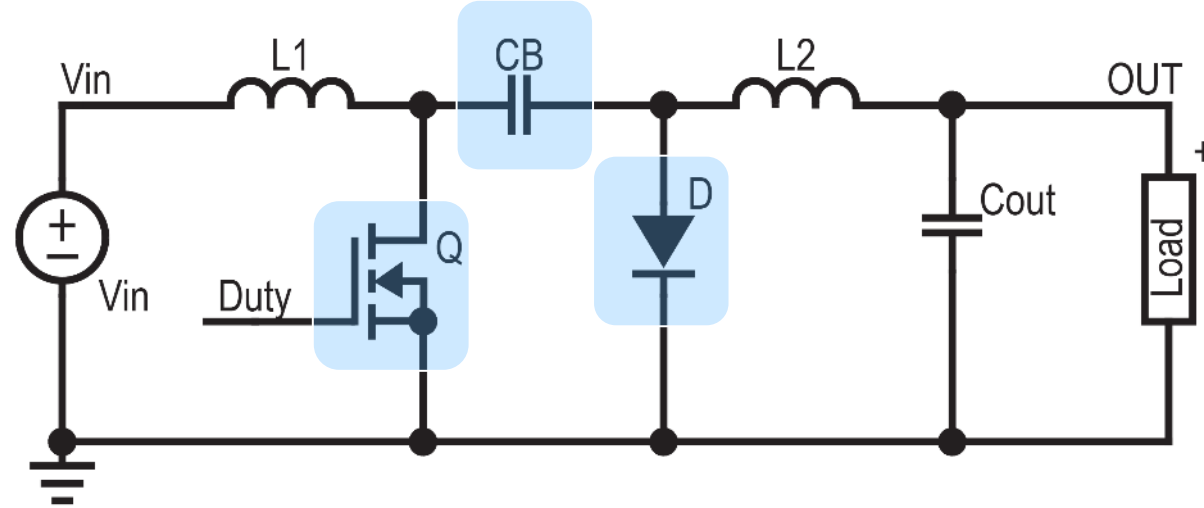
Cuk Converters



Cuk converters

- Inductors can be Coupled or Uncoupled
- Cap C_B provides DC isolation when the converter is OFF.
- Input and output current are continuous, making this a low-noise topology

Cuk Converters



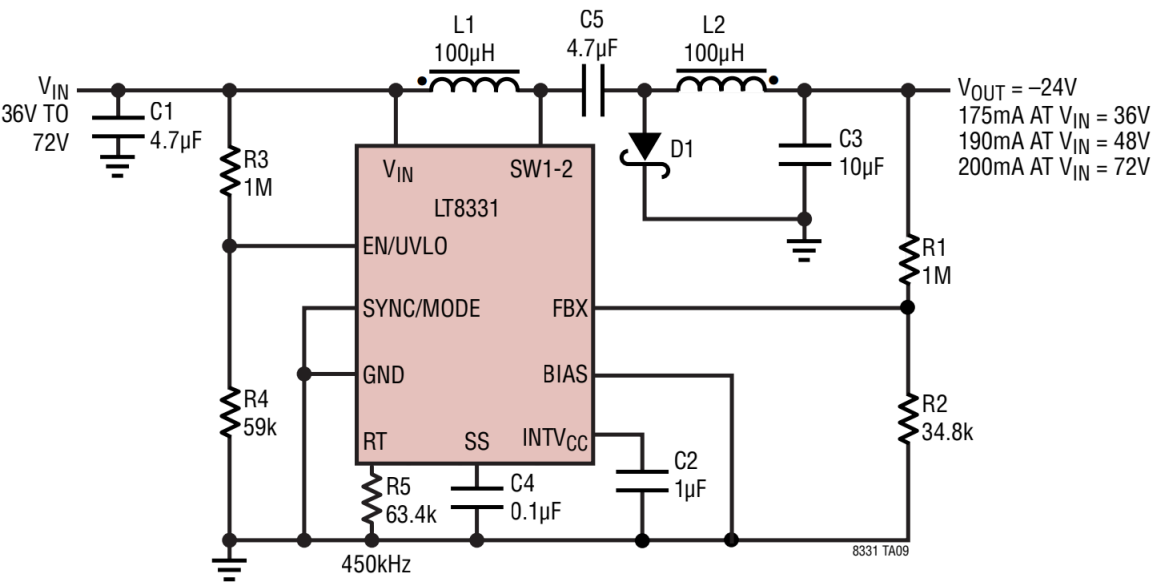
Cuk converters

- Voltage ratings:
 - Primary Switch will see $V_{IN} + |V_{OUT}|$
 - Coupling cap will see $V_{IN} + |V_{OUT}|$
 - Diode will see $V_{IN} + |V_{OUT}|$
- Current Ratings (average):
 - Primary Switch will see $(I_{IN} + I_{OUT})$
 - Diode will see I_{OUT}

Cuk Converters

Cuk converters

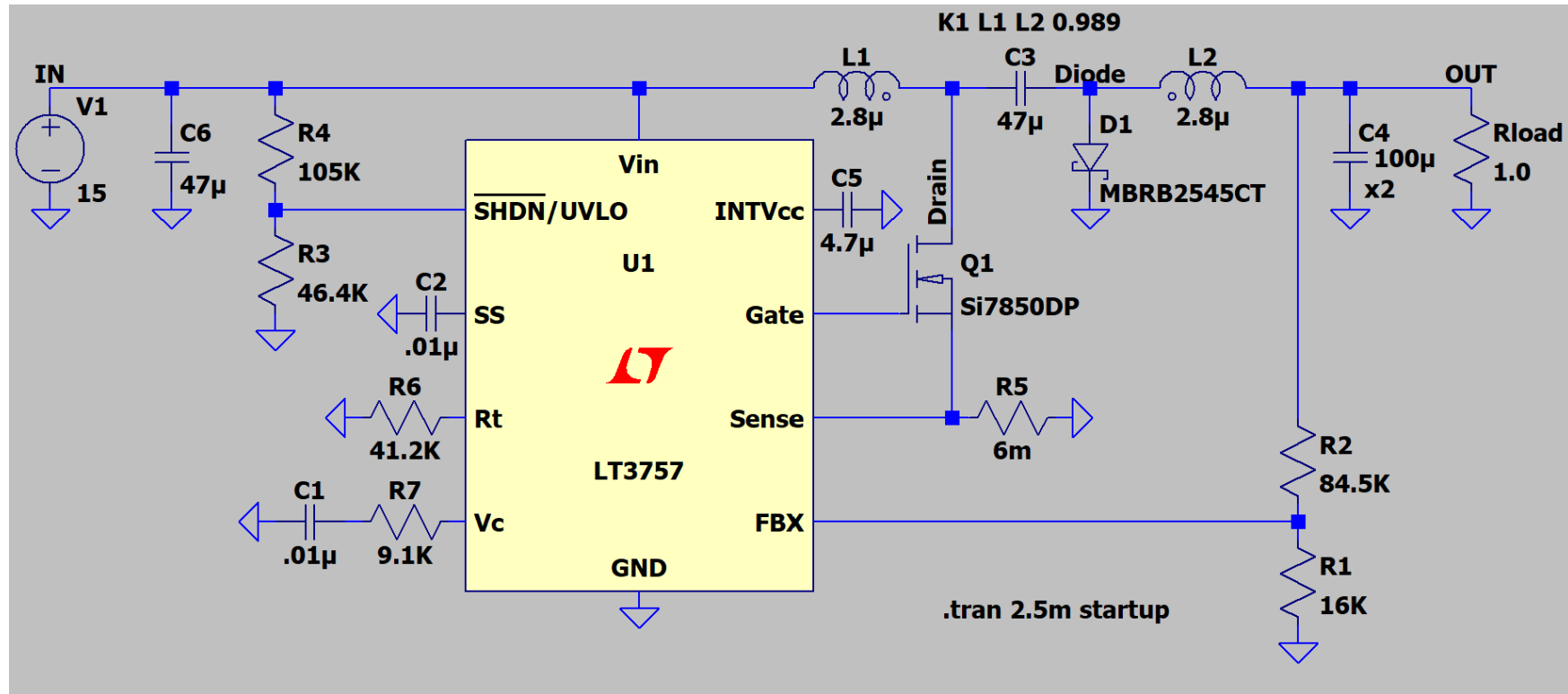
- FBX Pin
 - Several converters that are suitable for Cuk operation have a Bi-directional FB pin.
 - The FBX pin can switch automatically from regulating a positive output to a negative output
 - This simplifies setting V_{OUT} for Boost or Cuk topologies



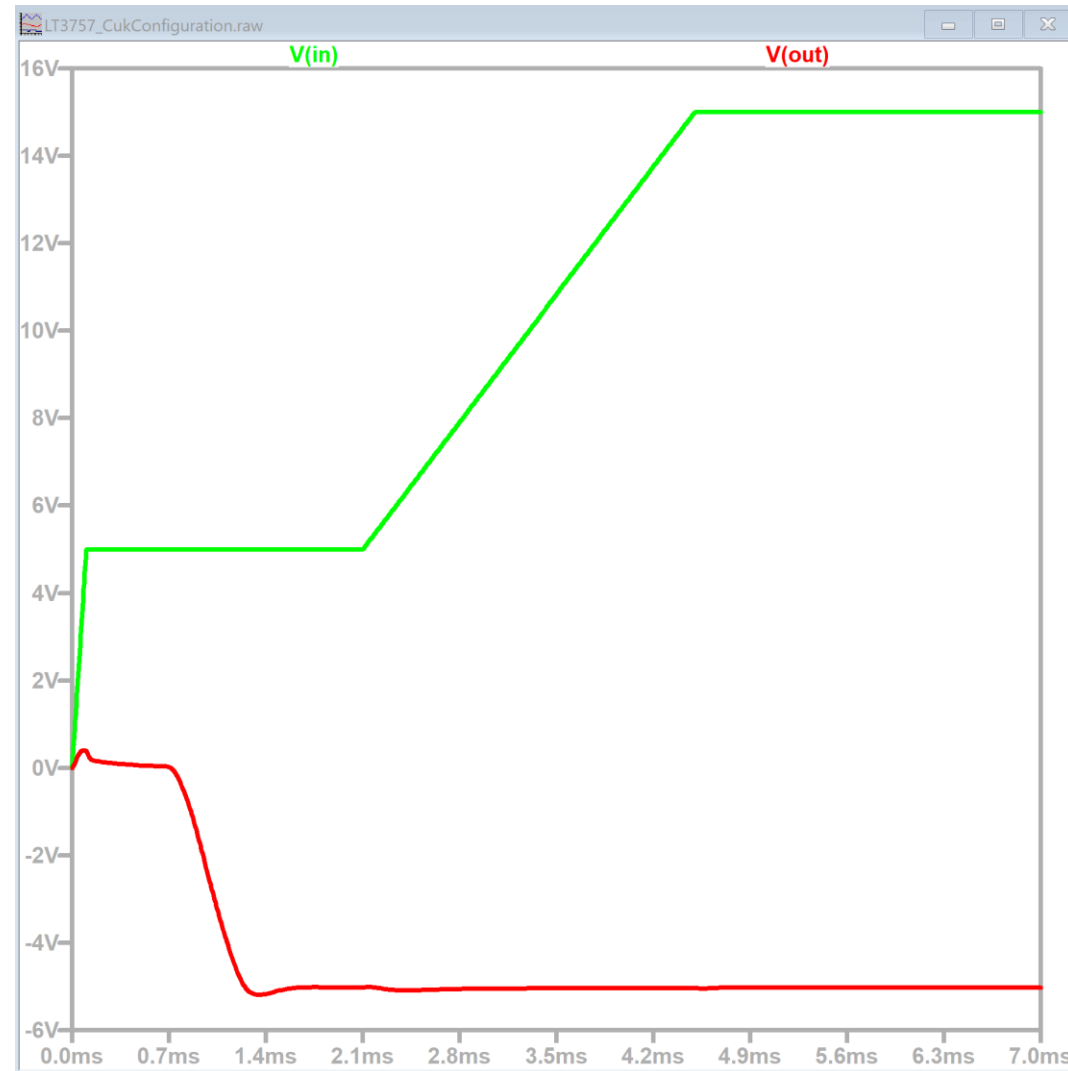
FBX Regulation						
FBX Regulation Voltage	FBX > 0V	●	1.568	1.6	1.632	V
	FBX < 0V	●	-0.820	-0.80	-0.780	V

Cuk Converter LTSPICE example

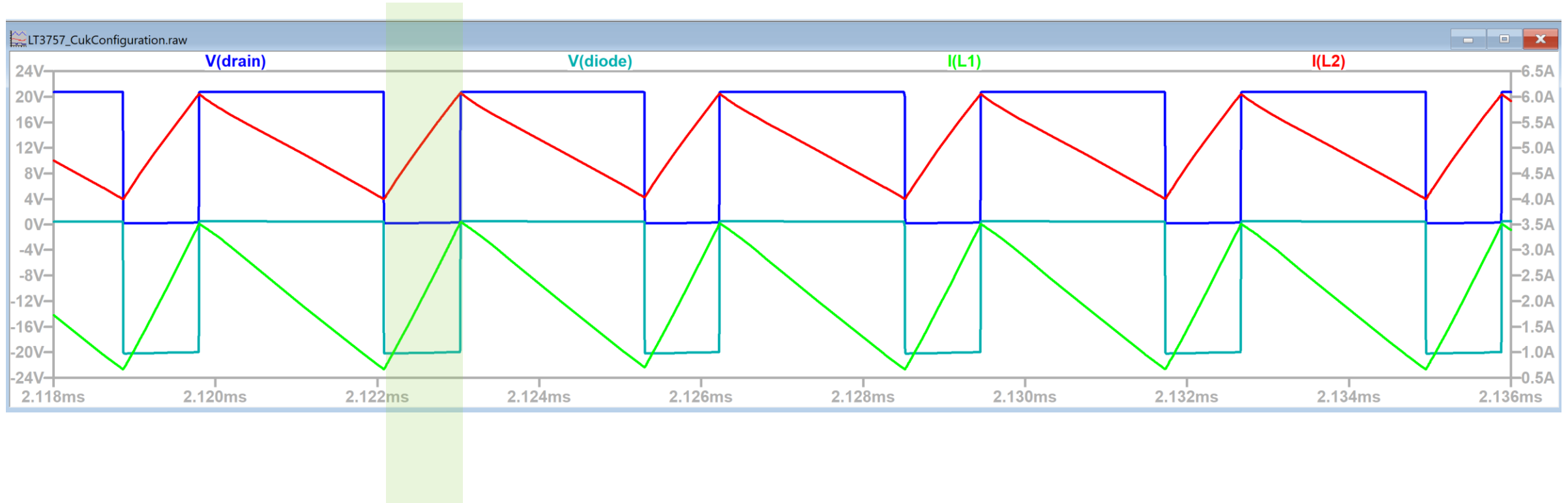
- LT3757 chosen as example
 - V_{in} : 5V to 15V
 - V_{out} : -5V, 3A to 5A at 300KHz



Cuk Converter LTSPICE example



Cuk Converter LTSPICE example



Cuk Converter LTSPICE example



Cuk Converters

Advantages

- Can generate negative V_{OUT} from a positive V_{IN}
- Provides output disconnect
- Input and output current are continuous – leading to low-noise V_{OUT}

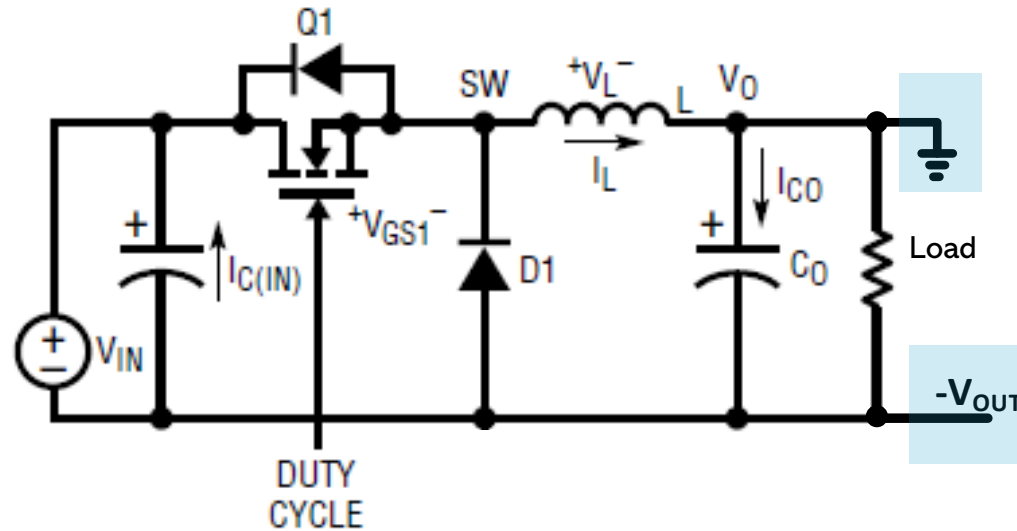
Disadvantages

- Component ratings (Voltage) are high
- Lower efficiency
- Requires 2 inductors

Agenda

- 1 Sepic
- 2 Ćuk Regulators
- 3 Inverting Regulators

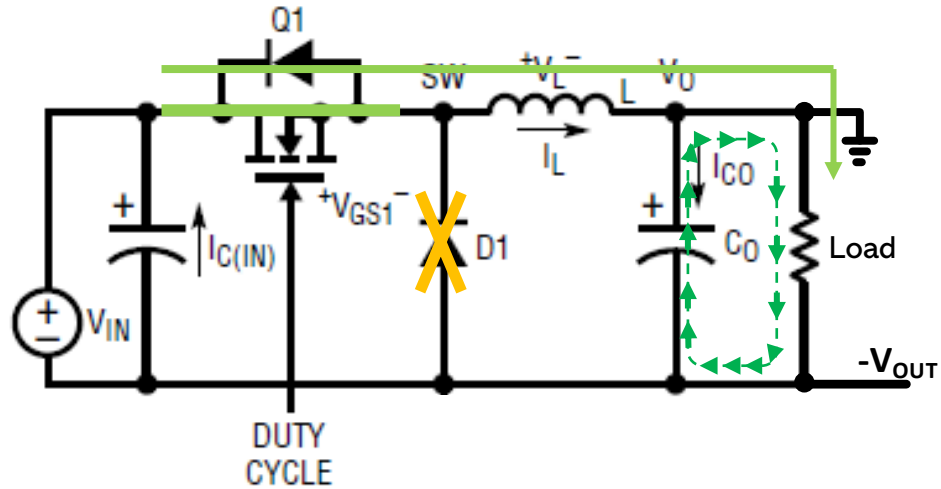
Inverting Converters



Buck-Boost Inverting Converters

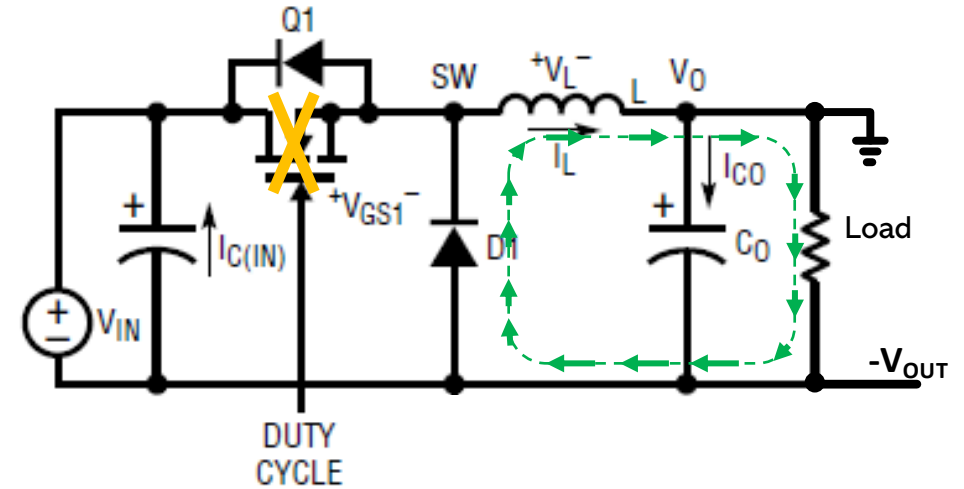
- The Ćuk converters are inverting by their nature.
- The Buck-Boost inverting topology is an alternate topology for generating a negative output.

Inverting Converters



ON-Phase:

- Primary switch feeds current through inductor to GND.
- Inductor current ramps up.
- Output current is provided by output cap.
- Note: Output current in discontinuous



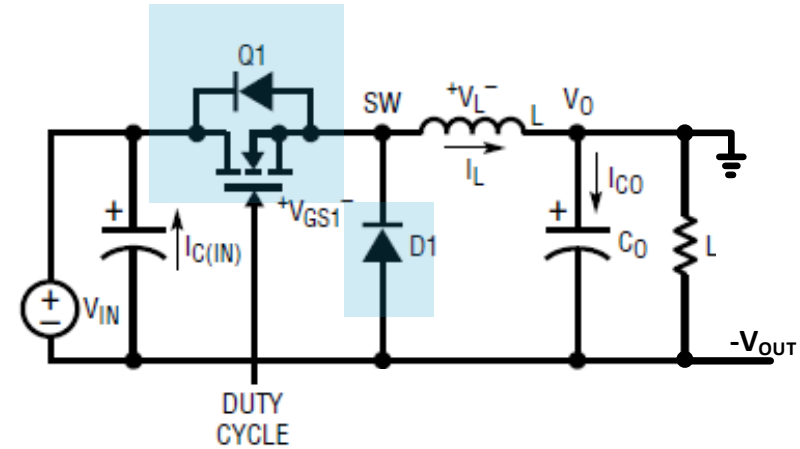
OFF-Phase:

- Primary switch is OFF.
- current flows through diode and inductor.
- Inductor current ramps down.

Inverting Converters

Buck-Boost Inverting Converters

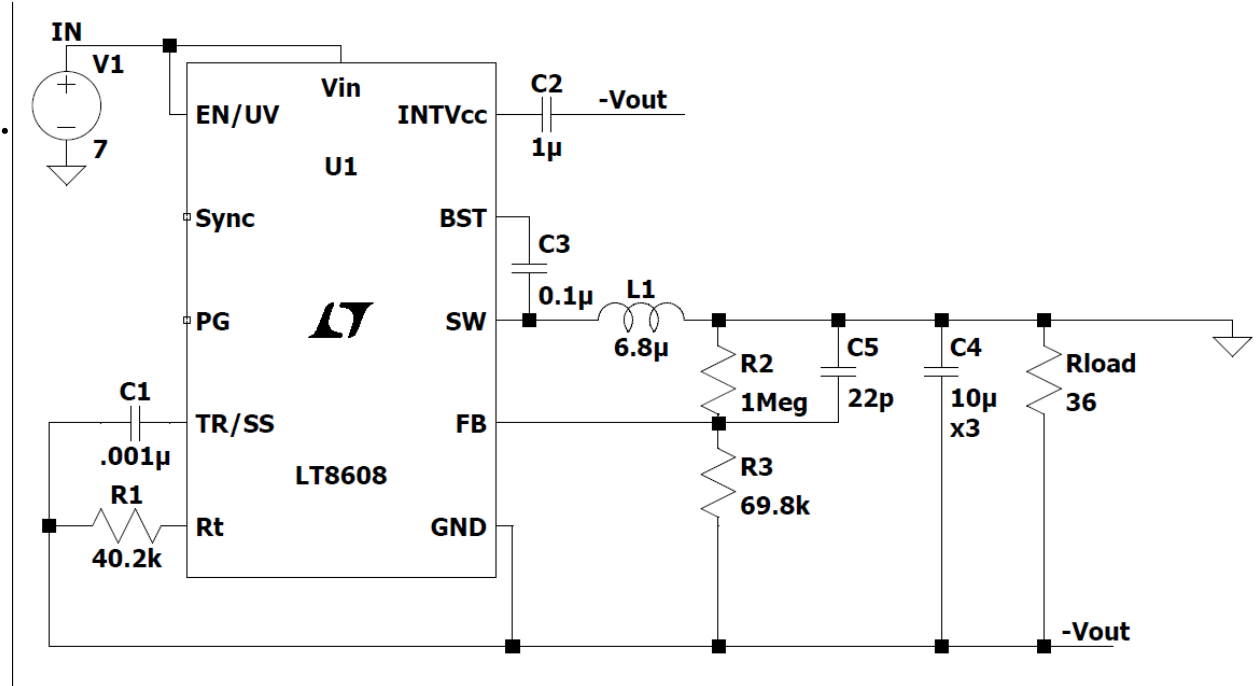
- Output current is discontinuous, leading to higher output ripple and noise.
- Voltage ratings:
 - Primary Switch will see $V_{IN} + |V_{OUT}|$
 - Diode will see $V_{IN} + |V_{OUT}|$
- Current Ratings (average):
 - Primary Switch will see $(I_{IN} + I_{OUT})$
 - Diode will see I_{OUT}



Inverting Converters

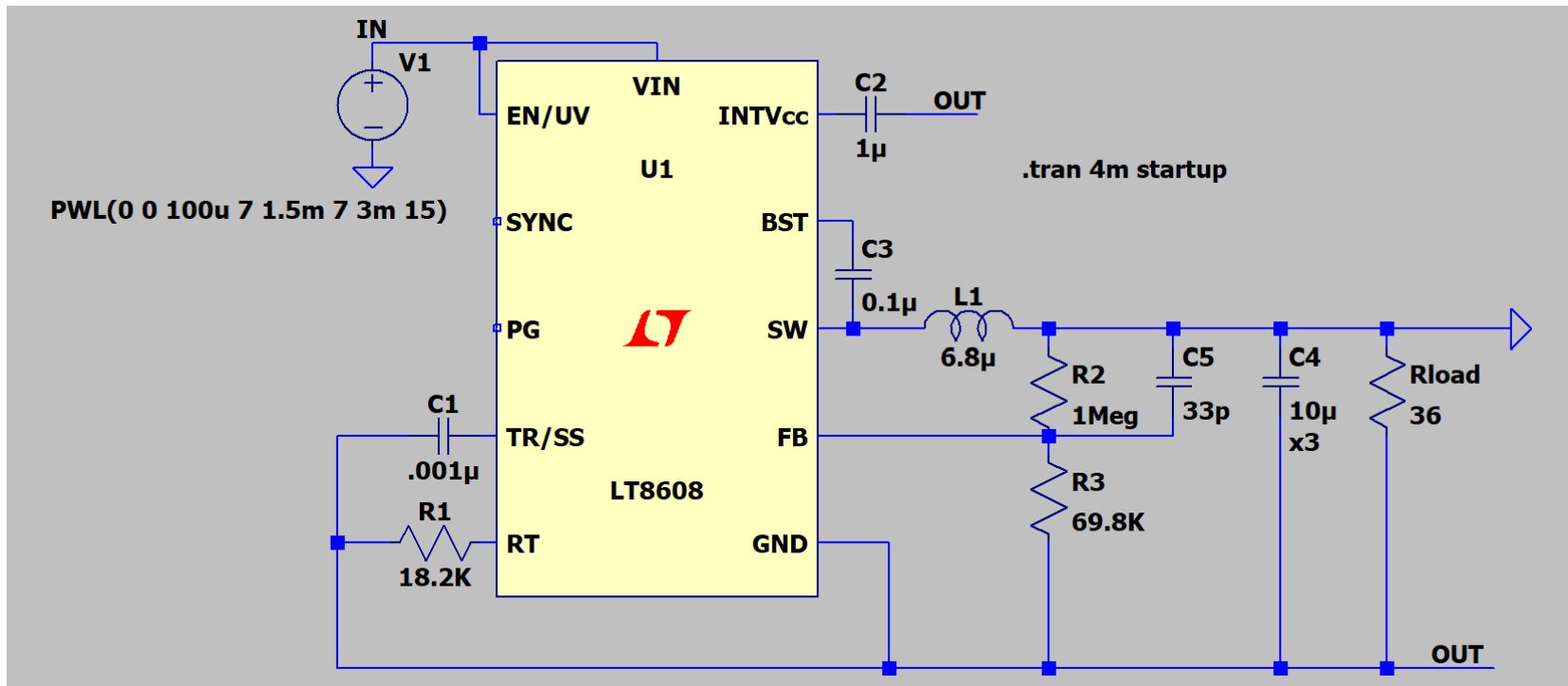
Buck-Boost Inverting Converters

- Can be created with any Buck converter (so there are lots of choices).
- Monolithic, controller, μ Module
- Note that IC GND pin is connected to $-V_{OUT}$. Control pins and FB divider are referenced to $-V_{OUT}$.



Buck-Boost Converter LTSPICE example

- LT8608 chosen as example
 - $V_{IN} = 7V$ to $15V$, $V_{OUT} = -12V$ @ $300mA$



LT8608/LT8608B

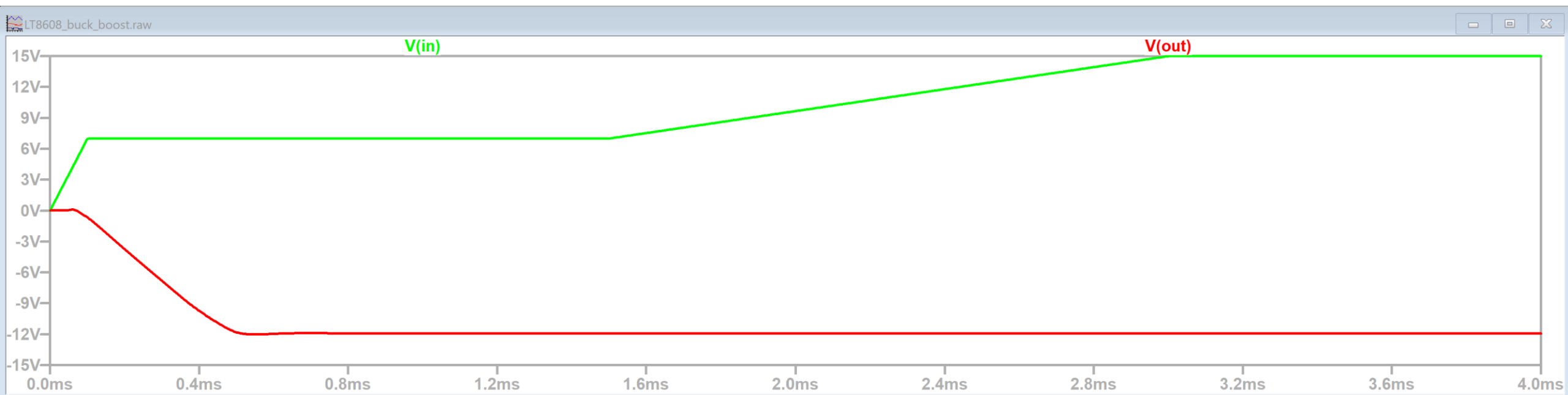
ABSOLUTE MAXIMUM RATINGS

(Note 1)

V_{IN} , EN/UV, PG.....	42V
FB, TR/SS	4V
SYNC Voltage	6V

Buck-Boost Converter LTSPICE example

- LT8608 chosen as example
 - $V_{IN} = 7V$ to $15V$, $V_{OUT} = -12V$ @ $300mA$



Inverting Converters

Buck-Boost Inverting Converters

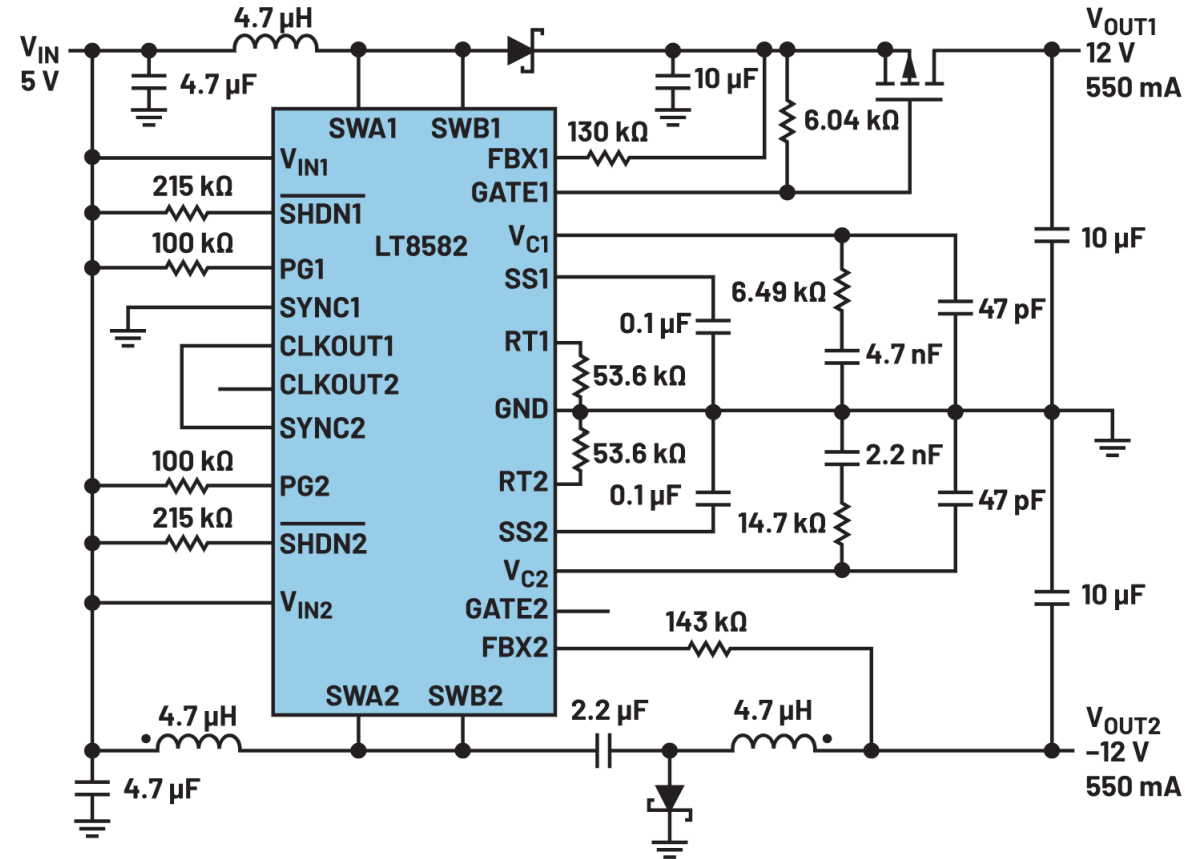
Advantages

- Can generate negative V_{OUT} from a positive V_{IN}
- Provides output disconnect
- Can use any Buck converter IC
- Good efficiency
- Single inductor

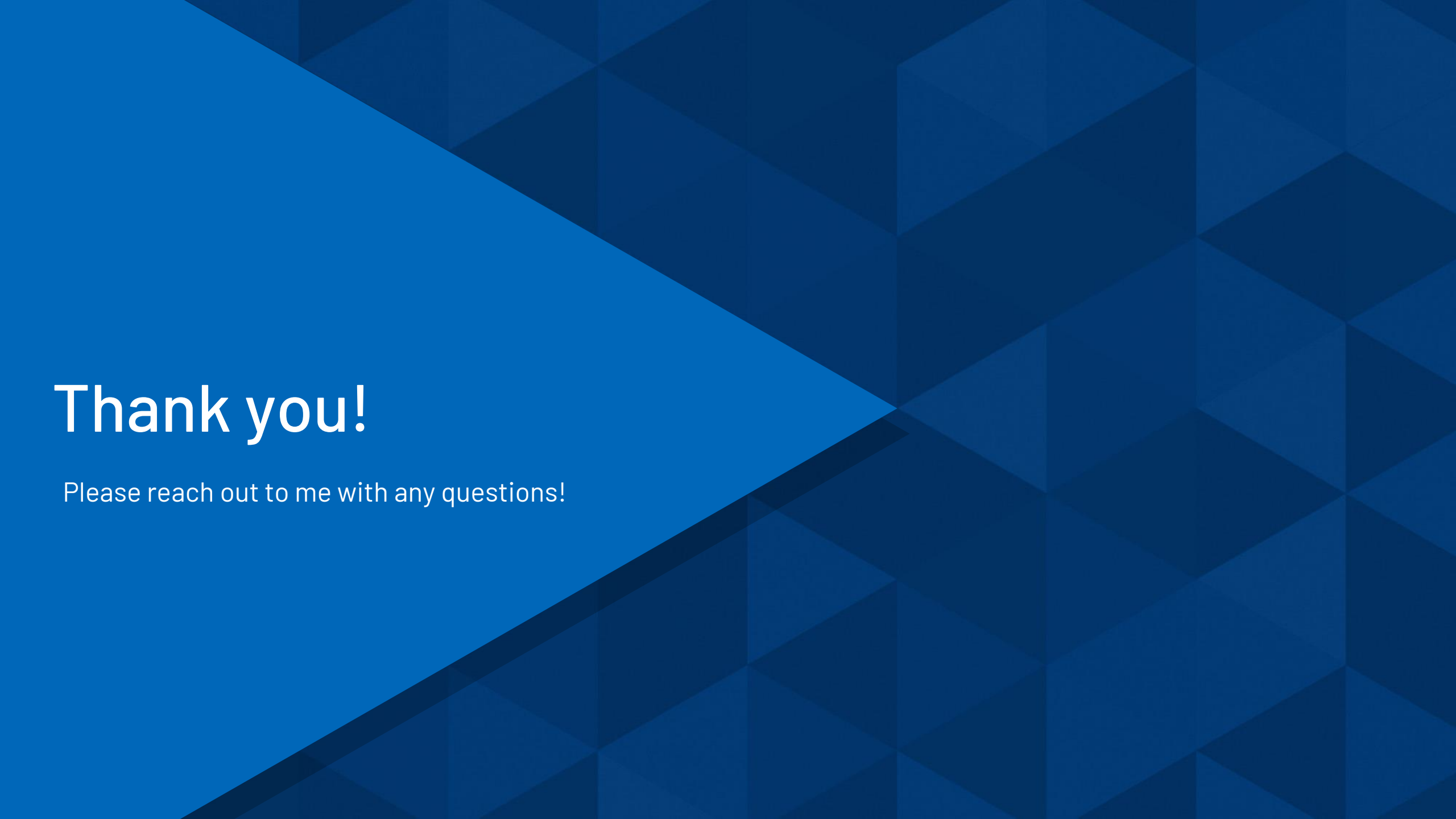
Disadvantages

- Component ratings (Voltage) are high
- FB pin, control pins are referenced to $-V_{OUT}$
- Output current is discontinuous

Dual Topology converter



- Dual 42 Vin, 3A boost/inverting regulator: [LT8582](#)
- Dual 50 Vin, 2A multi-topology regulator : [LT8471](#)
- Dual 5.5 Vin, 2A/1.2A boost/inverting regulator : [ADP5076](#)



Thank you!

Please reach out to me with any questions!

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